EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 5.01

C Circle Theorems, Intersecting Chords & Mensuration

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

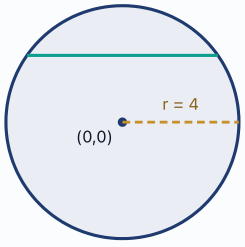
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 C	Q4 D	Q5 A
Q6 B	Q7 C	Q8 D	Q9 B	Q10 D
Q11 B	Q12 C	Q13 C	Q14 B	Q15 C
Q16 B	Q17 A	Q18 D	Q19 C	Q20 A
Q21 A	Q22 B	Q23 B	Q24 A	Q25 C

Circle Theorems, Intersecting Chords & Mensuration Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 14x + 64$ in the domain of Circle Theorems, Similar Solids & Mensuration. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(9)$.



A 34 ✓

B 41

C 19

D 32

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Circle Theorems, Similar Solids & Mensuration):

1. Complete the square: $x^2 - 14x + 64 = (x - 7)^2 - 49 + 64 = (x - 7)^2 + 15$, giving minimum value $k = 15$. **[M1]**
2. Evaluate at $x = 9$: $f_1(9) = (2)^2 + 15 = 19$. **[M1]**
3. Sum the two quantities: $k + f_1(9) = 15 + 19 = 34$. **[A1]**

- **Final Verified Answer:** 34.

2 MCQ

A mathematical model in Circle Theorems, Similar Solids & Mensuration is defined by the cubic function $y_2(x) = 3x^3 - 7x^2 + 15x - 12$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 96

B 54 ✓

C 39

D 28

WORKED SOLUTION & VERIFICATION

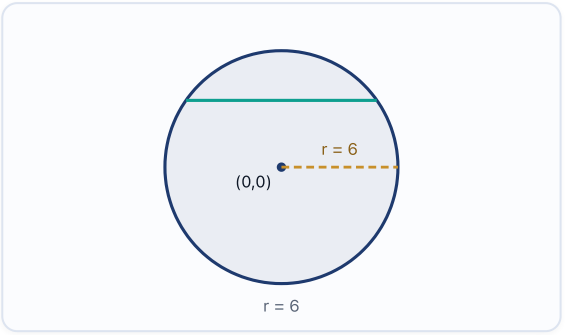
Mark Scheme & Examiner Solution (Circle Theorems, Similar Solids & Mensuration):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 14x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 9(9) - 14(3) + 15 = 54$. **[M1][A1]**

- **Final Verified Answer:** 54.

3 MCQ

In Circle Theorems, Similar Solids & Mensuration, two displacement vectors are given by $\mathbf{u}_3 = \langle 7, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 7

B 113

C 120 ✓

D 232

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Circle Theorems, Similar Solids & Mensuration):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (7)(9) + (-8)(7) = 63 + (-56) = 7$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 7^2 + (-8)^2 = 113$. **[M1]**
3. Combine: $7 + 113 = 120$. **[A1]**

- **Final Verified Answer:** 120.

4 MCQ

The linear growth profile shown in the diagram for Circle Theorems, Similar Solids & Mensuration follows an arithmetic progression with first term $a_1 = 8$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 92

B 840

C 1380

D 750 ✓

WORKED SOLUTION & VERIFICATION

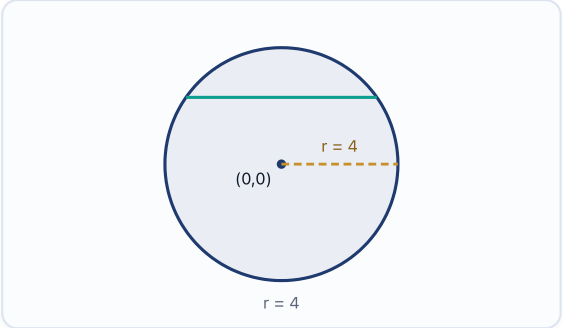
Mark Scheme & Examiner Solution (Circle Theorems, Similar Solids & Mensuration):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 8, d = 6$: $S_{15} = \frac{15}{2}[16 + 14 \times 6] = \frac{15}{2} \times 100 = 750$. **[M1][A1]**

- **Final Verified Answer:** 750.

5 MCQ

In Circle Theorems, Similar Solids & Mensuration, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 28x)\,dx$.



A 104 ✓

B 128

C 132

D 53

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Circle Theorems, Similar Solids & Mensuration):

1. Integrate term by term: $\int (18x^2 + 28x)\,dx = [6x^3 + 14x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 14(2)^2) - 0 = 48 + 56 = 104$. **[M1][A1]**

- **Final Verified Answer:** 104.

6 MCQ

In Circle Theorems, Similar Solids & Mensuration, consider the quadratic function $f_6(x) = x^2 - 16x + 151$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(15)$.

A 231

B 223 ✓

C 136

D 216

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Circle Theorems, Similar Solids & Mensuration):

1. Complete the square: $x^2 - 16x + 151 = (x-8)^2 - 64 + 151 = (x-8)^2 + 87$, giving minimum value $k = 87$. **[M1]**
2. Evaluate at $x = 15$: $f_6(15) = (7)^2 + 87 = 136$. **[M1]**
3. Sum the two quantities: $k + f_6(15) = 87 + 136 = 223$. **[A1]**

- **Final Verified Answer:** 223.

7 MCQ

The graph illustrates the cubic model $y_{7}(x) = 3x^{3} - 8x^{2} + 33x - 19$ studied in Circle Theorems, Similar Solids & Mensuration. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 609

B 448

C 481 ✓

D 89

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Circle Theorems, Similar Solids & Mensuration):

1. Differentiate term by term using the power rule: $\frac{\mathrm{d}y}{\mathrm{d}x} = 9x^{2} - 16x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=8} = 9(64) - 16(8) + 33 = 481$. **[M1][A1]**

- **Final Verified Answer:** 481.

8 MCQ

In Circle Theorems, Similar Solids & Mensuration, two displacement vectors are given by $\mathbf{u}_{8} = \langle 8, -7 \rangle$ and $\mathbf{v}_{8} = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_{8} \cdot \mathbf{v}_{8} + \|\mathbf{u}_{8}\|^{2}$.

A 29

B 113

C 212

D 142 ✓

WORKED SOLUTION & VERIFICATION

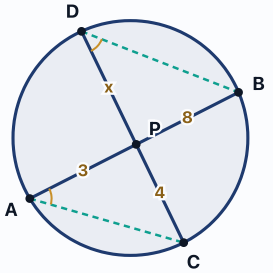
Mark Scheme & Examiner Solution (Circle Theorems, Similar Solids & Mensuration):

1. Compute the inner product: $\mathbf{u}_{8} \cdot \mathbf{v}_{8} = (8)(8) + (-7)(5) = 64 + (-35) = 29$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_{8}\|^{2} = 8^{2} + (-7)^{2} = 113$. **[M1]**
3. Combine: $29 + 113 = 142$. **[A1]**

- **Final Verified Answer:** 142.

9 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 3$ cm, $PB = 8$ cm, and $PC = 4$ cm, find the length PD.



A 5 cm

B 6 cm ✓

C 9 cm

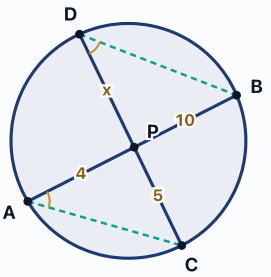
D 7 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 3 \times 8 = 4 \cdot PD \implies PD = 6$ cm.

10 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 4$ cm, $PB = 10$ cm, and $PC = 5$ cm, find the length PD.



A 9 cm

B 11 cm

C 6 cm

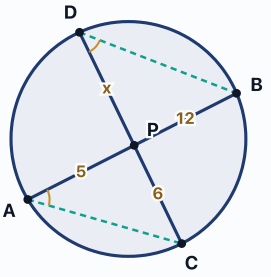
D 8 cm ✓

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 4 \times 10 = 5 \cdot PD \implies PD = 8$ cm.

11 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 5$ cm, $PB = 12$ cm, and $PC = 6$ cm, find the length PD.



A 7 cm

B 10 cm ✓

C 11 cm

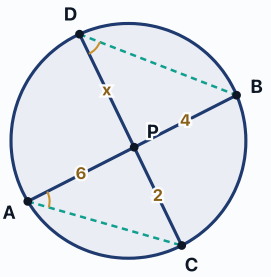
D 13 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 5 \times 12 = 6 \cdot PD \implies PD = 10$ cm.

12 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 6$ cm, $PB = 4$ cm, and $PC = 2$ cm, find the length PD.



A 8 cm

B 1 cm

C 12 cm ✓

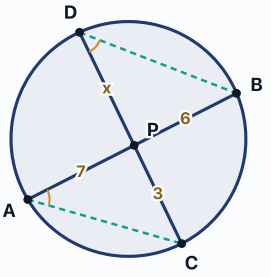
D 15 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 6 \times 4 = 2 \cdot PD \implies PD = 12$ cm.

13 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 7$ cm, $PB = 6$ cm, and $PC = 3$ cm, find the length PD.



A 10 cm

B 17 cm

C 14 cm ✓

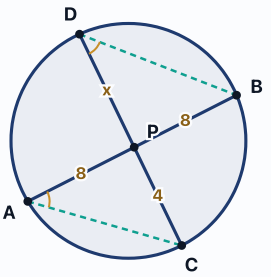
D 1 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 7 \times 6 = 3 \cdot PD \implies PD = 14$ cm.

14 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 8$ cm, $PB = 8$ cm, and $PC = 4$ cm, find the length PD.



A 1 cm

B 16 cm ✓

C 12 cm

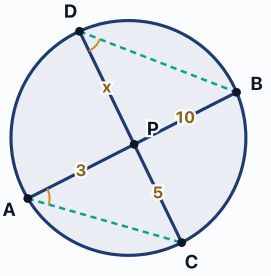
D 19 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 8 \times 8 = 4 \cdot PD \implies PD = 16$ cm.

15 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 3$ cm, $PB = 10$ cm, and $PC = 5$ cm, find the length PD.



A 8 cm

B 9 cm

C 6 cm ✓

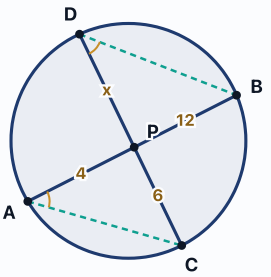
D 7 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 3 \times 10 = 5 \cdot PD \implies PD = 6$ cm.

16 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 4$ cm, $PB = 12$ cm, and $PC = 6$ cm, find the length PD.



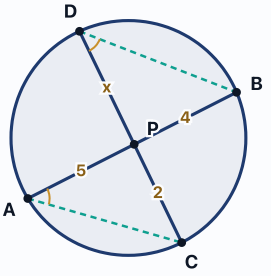
A 10 cm	B 8 cm ✓
C 11 cm	D 14

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 4 \times 12 = 6 \cdot PD \implies PD = 8$ cm.

17 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 5$ cm, $PB = 4$ cm, and $PC = 2$ cm, find the length PD.



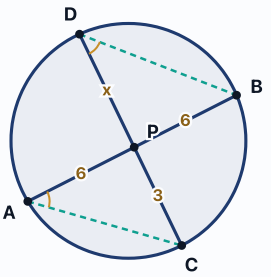
A 10 cm ✓	B 1 cm
C 7 cm	D 13 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 5 \times 4 = 2 \cdot PD \implies PD = 10$ cm.

18 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 6$ cm, $PB = 6$ cm, and $PC = 3$ cm, find the length PD.



A 15 cm

B 9 cm

C 1 cm

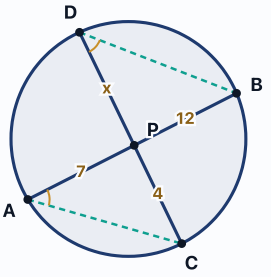
D 12 cm ✓

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 6 \times 6 = 3 \cdot PD \implies PD = 12$ cm.

19 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 7$ cm, $PB = 12$ cm, and $PC = 4$ cm, find the length PD.



A 24 cm

B 15 cm

C 21 cm ✓

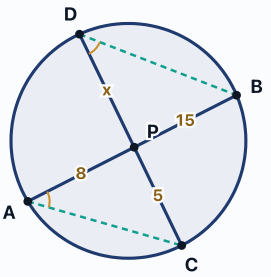
D 5 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 7 \times 12 = 4 \cdot PD \implies PD = 21$ cm.

20 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 8$ cm, $PB = 15$ cm, and $PC = 5$ cm, find the length PD.



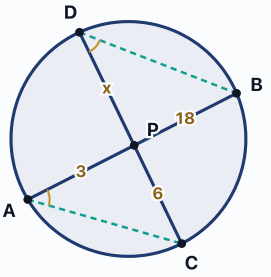
A 24 cm ✓	**B** 27 cm
C 7 cm	**D** 18 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 8 \times 15 = 5 \cdot PD \implies PD = 24$ cm.

21 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 3$ cm, $PB = 18$ cm, and $PC = 6$ cm, find the length PD.



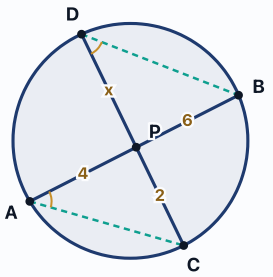
A 9 cm ✓	**B** 14
C 15 cm	**D** 12 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 3 \times 18 = 6 \cdot PD \implies PD = 9$ cm.

22 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 4$ cm, $PB = 6$ cm, and $PC = 2$ cm, find the length PD.



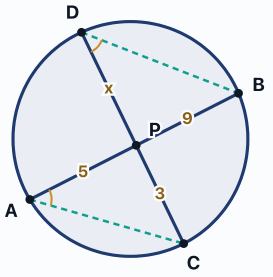
A 8 cm	B 12 cm ✓
C 2 cm	D 15 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 4 \times 6 = 2 \cdot PD \implies PD = 12$ cm.

23 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 5$ cm, $PB = 9$ cm, and $PC = 3$ cm, find the length PD.



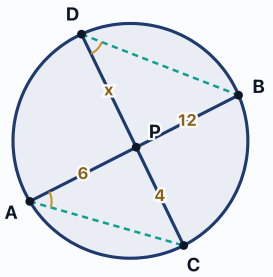
A 4 cm	B 15 cm ✓
C 18 cm	D 11 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 5 \times 9 = 3 \cdot PD \implies PD = 15$ cm.

24 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 6$ cm, $PB = 12$ cm, and $PC = 4$ cm, find the length PD.



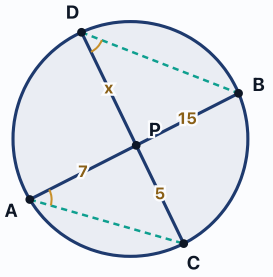
- **A** 18 cm ✓
- **B** 14 cm
- **C** 21 cm
- **D** 6 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 6 \times 12 = 4 \cdot PD \implies PD = 18$ cm.

25 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 7$ cm, $PB = 15$ cm, and $PC = 5$ cm, find the length PD.



- **A** 24 cm
- **B** 17 cm
- **C** 21 cm ✓
- **D** 8 cm

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem, $PA \cdot PB = PC \cdot PD \implies 7 \times 15 = 5 \cdot PD \implies PD = 21$ cm.